Arduino-based Experiment Lab Notebook[†]

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1 Abstract

This experiment explores Arduino-based automation for controlling electronic and mechanical systems. It involves programming an Arduino to operate an LED with variable brightness and a stepper motor for linear motion. The stepper motor is then used to drive a sensor for measuring the axial variation of the magnetic field strength of a coil. The study includes analyzing the resolution of motion control, backlash error, and the effects of ADC digitization on measured field profiles.

2 Aim

To program and utilize an Arduino for controlling an LED and a stepper motor, and to measure the axial variation of a magnetic field using a stepper motor-driven sensor while analyzing system accuracy and resolution.

3 Apparatus

- Arduino and arduino IDE
- Breadboard and wires
- Resistors
- Stepper Motor
- Rheostat
- LED
- Magnetic field sensor (Hall effect sensor)
- Electromagnetic coil
- Ruler/Caliper

[†]Use Digital copy for the colored plots and diagrams.

4 Theory

Arduino is an open-source electronics platform based on easy-to-use hardware and software. It consists of a programmable microcontroller board that can read inputs from sensors, process data, and control outputs such as LEDs, motors, and displays. It is widely used for prototyping, automation, and embedded system development.

subsectionHall Effect

Hall effect occurs when a current-carrying conductor or semiconductor is placed in a perpendicular magnetic field. The interaction between the charge carriers and the magnetic field leads to the development of a transverse voltage, known as the Hall voltage.

Working principle: Moving charges inside a conductor experience a Lorentz force due to the magnetic field, accumulating charges on one side of the conductor. This creates an electric field that opposes the magnetic force.

The Lorentz force acting on a charge carrier q moving with drift velocity v_d is given by:

$$\mathbf{F} = q(\mathbf{E} + \mathbf{v_d} \times \mathbf{B})$$

In equilibrium, the electric force due to the Hall voltage V_H balances the Lorentz force:

$$qE_H = qv_dB$$

The Hall electric field is related to the Hall voltage by:

$$E_H = \frac{V_H}{W}$$

where W is the width of the conductor, giving:

$$V_H = v_dBW$$

Using the relation between drift velocity and current density J:

$$J = nqv_d$$

where n is the charge carrier density, we derive the **Hall voltage**:

$$V_H = \frac{IB}{nqW}$$

The *Hall coefficient* R_H is defined as:

$$R_H = \frac{E_H}{JB} = \frac{1}{nq}$$

4.1 Magnetic Field of Solenoid

For an ideal solenoid of length L, radius R, and total number of turns N carrying current I, the number of turns per unit length is:

$$n = \frac{N}{L}$$

Using Ampere's circuital law:

$$\oint \mathbf{B} \cdot d\mathbf{l} = \mu_0 I_{enc}$$

The magnetic field inside the solenoid is:

$$B = \mu_0 n I = \mu_0 \frac{N}{L} I$$

For a finite solenoid, the magnetic field at a point on its central axis at distance z from the center is:

$$B = \frac{\mu_0 n I}{2} \left[\frac{z_2}{\sqrt{R^2 + z_2^2}} - \frac{z_1}{\sqrt{R^2 + z_1^2}} \right]$$

or equivalently:

$$B = \frac{\mu_0 n I}{2} [\cos(\theta_1) + \cos(\theta_2)]$$

4.2 LED Control Using Arduino

4.2.1 Principle of Operation:

The brightness of an LED is controlled using Pulse Width Modulation (PWM), where the duty cycle of a digital signal determines the average voltage supplied to the LED.

4.2.2 Working Principle:

- The Arduino generates a PWM signal that controls the LED's brightness by rapidly switching it on and off.
- By varying the duty cycle, the perceived brightness changes.
- A timer can be programmed to automate LED operation.

4.3 Stepper Motor Control and Backlash Analysis

4.3.1 Principle of Operation:

A Stepper Motor moves in discrete steps by energizing coils in a sequential manner. The step size and resolution depend on the step angle and microstepping.

4.3.2 Working Principle:

- The Arduino sends signals to the stepper motor driver to rotate the motor in steps.
- The motor moves a carriage forward and backward on a linear axis.

- The resolution of motion control is studied by analyzing the smallest detectable movement.
- Backlash error (the difference between expected and actual motion) is measured by comparing forward and backward movements over different distances.

4.4 Magnetic Field Measurement Using Stepper Motor-Driven Sensor

4.4.1 Principle of Operation:

The Magnetic Field Sensor (e.g., Hall effect sensor) detects the strength of a magnetic field based on the voltage induced by the interaction of the field with charge carriers in the sensor material.

4.4.2 Working Principle:

- The stepper motor moves the sensor along the axis of an electromagnetic coil.
- The sensor measures the magnetic field at different positions.
- The Arduino records the data and plots the field variation for different coil currents.
- The Arduino records the data and plots the field variation for different coil currents.
- The resolution of the stepper motor and the digitization effect of the ADC on the measured field are analyzed.

5 Diagram of the Arduino UNO Board

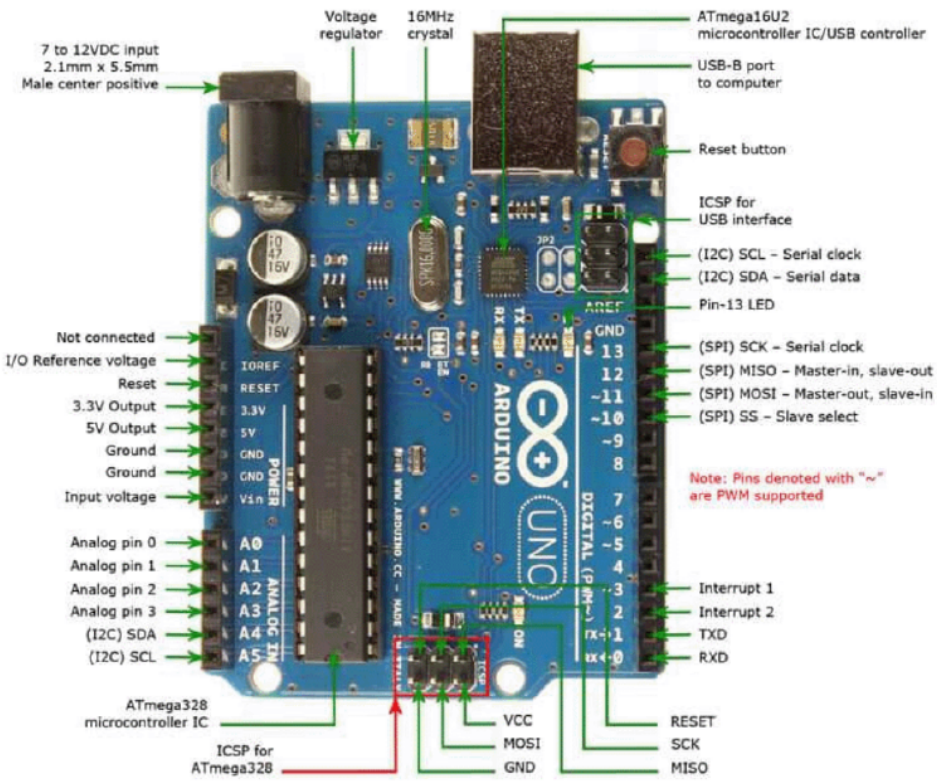


Figure 1: Arduino UNO Board

6 Experimental Codes

6.1 Activity 1

We used the following two codes to set a timer for operating the LED in on or off state and to control the LED's brightness.

```
void setup()
{
  pinMode(10, OUTPUT);
}

void loop()
{
  digitalWrite(10, HIGH);
  delay(1000);
  digitalWrite(10, LOW);
  delay(1000);
}
```

Figure 2: Code to set a time to operate the LED

```
void setup()
{
  pinMode(10, OUTPUT);
}

void loop()
{
  for(int i = 0; i<=255; i++)
  {
  analogWrite(10, i);
  delay(100);
  }
  for(int i=255; i>=0; i--)
  {
  analogWrite(10, i);
  delay(100);
  }
  exit(0);
}
```

Figure 3: Code to control the LED's brightness

6.2 Activity 2

For activity 2, we used the following code to let the arduino control the stepper-motor to make a carriage move forward and backward on a linear axis using which the resolution and the backlash error if any studied.

```
int DIR = 8;
int PUL = 10;

void setup()
{
  pinMode(DIR, OUTPUT);
  pinMode(PUL, OUTPUT);
}

void loop()
{
  digitalWrite(DIR, HIGH);
  for (int i =0; i<200; i++)
  {
    digitalWrite(PUL, HIGH);
    delay(10);
    digitalWrite(PUL, LOW);
    delay(10);
  }
  exit(0);
}
```

Figure 4: Code to control the stepper-motor to move backward and forward

6.3 Activity 3

For activity 3, we used the following code to determine the axial variation of magnetic field strength using a Arduino-controlled stepper-motor driven sensor. Readings are taken for every 25 steps, and for each step 100 readings are calculated and averaged.

```
const int AnalogPin = A0;
const float GAUSS_PER_STEP = 2.52;
const float zeroLevel = 514;

float rawValue = 0.0;
float gaussValue = 0.0;

void setup() {
    pinMode(8, OUTPUT);
    pinMode(10, OUTPUT);
    Serial.begin(9600);
}

void loop() {
    float gauss = 0.0;
    for (int j = 25; j > 0; j--)
    {
        digitalWrite(8, HIGH);
        digitalWrite(10, LOW);
        delay(5);
        digitalWrite(10, HIGH);
    }

    delay(100);

    for (int i = 0; i < 100; i++) {
        rawValue = analogRead(AnalogPin) - zeroLevel;
        gaussValue = rawValue * GAUSS_PER_STEP;
        gauss += gaussValue;
        delay(10);
    }
    Serial.println(gauss / 100);
}
```

Figure 5: Code to determine the axial variation of magnetic field strength

7 Experimental data and Analysis

7.1 Activity 1

By running the 1st code we see that LED starts blinking for every 1 sec (because of delay(1000)). For 1 sec LED is ON and for next 1 sec LED is OFF.

And running the 2nd code, we see that at PWM pin(analogpin) 10, every 100 ms LED's brightness decreases or increases.

7.2 Activity 2

Data to find the resolution of the stepper-motor -

No. of steps	Initial Reading	Final Reading	Resolution(mm/steps)
200	68.44	76.4	0.0398
400	76.4	84.46	0.0403
600	84.46	92.58	0.0406
800	92.58	100.42	0.0392
1000	100.42	108.44	0.0401
1200	108.44	116.44	0.04
1400	116.44	124.52	0.0404
1600	124.52	132.5	0.0399

Table 1: Observation to calculate resolution of the stepper-motor

So the mean resolution $= \frac{0.0398+0.0403+0.0406+0.0392+0.0401+0.04+0.0404+0.0399}{8} = 0.04$ mm/steps

Data to find the backlash error of the stepper-motor -

No. of steps	Initial Reading	Final Reading	Backlash Error (in mm)
200	58.44	58.34	0.1
400	58.32	58.42	0.1
600	58.42	58.48	0.06
800	58.48	58.44	0.04
1000	58.44	58.42	0.02

Table 2: Observation to backlash error of the stepper-motor

So the mean backlash error is $= \frac{0.1+0.1+0.06+0.04+0.02}{5} = 0.064$ mm

7.3 Activity 3

Here is the plot of the axial variation of the magnetic field. We have taken readings by moving the stepper motor backwards and forwards (i.e. 2 times for each current).

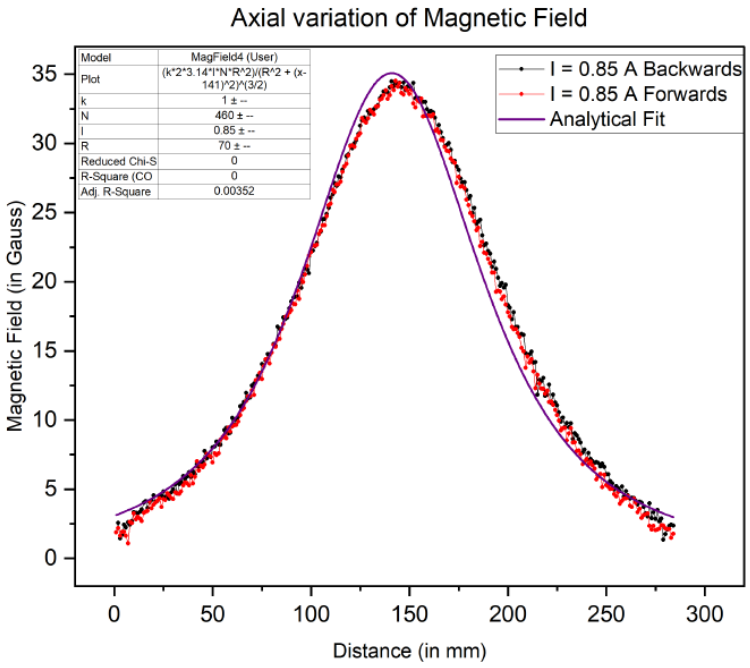


Figure 6: Plot of axial variation of magnetic field strength for I = 0.85 A

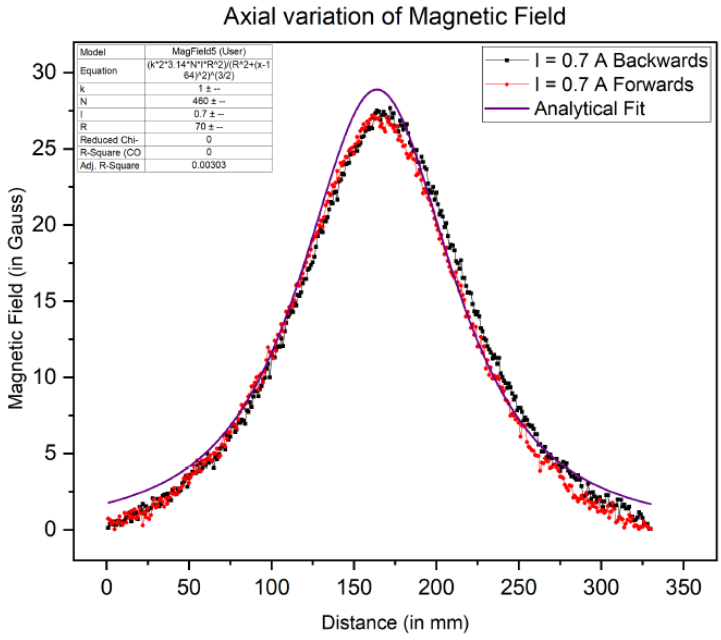


Figure 7: Plot of axial variation of magnetic field strength for I = 0.7 A

All these plots are done using putty logs using that model code.

8 Conclusions and Inferences

8.1 Activity 1:

8.1.1 LED blinking

- Code succesfully programmed on Arduino to blink an LED on and off at 1 sec delay by using digitalWrite function
- This experiment serves as an introduction and basic understanding to arduino programming and digital output manipulation.

8.1.2 LED brightness control

- Used PWM (Pulse Width Modulation) technique to control the brightness of the LED by using the duty cycle of the PWm signal using analogWrite() function
- We learned how to control analog output using PWM. This experiment demonstrates the ability to control analog output using PWM. This experiment demonstrates the ability to control LED brightness continuously using Arduino.

8.2 Activity 2

- We succesfully controlled the movemnet of the stepper-motor using arduino board. We move the stepper-motor with a interval of 200 steps and for every 200 steps we note down the reading and find the resolution of the stepper-motor which is 0.04 mm/steps.
- Next we studied the backlash error by the to-and-fro motion of the stepper-motor. We find that the backlash error decreases with increase in step size. The mean backlash error calculated from the data is 0.064 mm.

8.3 Activity 3

Integrated a hall effect sensor with a stepper motor system to detect the magnetic field generated by a 460 N turn and 7 cm coil with different currents. Recorded the number of steps taken by the motor for a 25-25 difference.

- As we demonstrated the ability to interface for a sensor with arduino to provide feedback for closed loop control. Analysed the relationship between the magnetic field and stepper motor steps (i.e. current and distance from the center of the coil).
- The magnetic field, using analytical method $B = \frac{u_o}{4\pi} \frac{2\pi N I R^2}{(R^2+x^2)^{3/2}}$, where B = magnetic field of the coil and N = 460 and R is 70 mm (Radius of the coil).

  We took the reading and plot the graph in the origin and fit the curve using same formula.
- From the Figure 6 and 7, we can see that the field is maximum at the center of the coil and decreases as we move outwards from the center.

9 Precautions and Sources of error

- Stepper motor backlash leading to positional inaccuracies.
- Provide sufficient power for the stepper motor to ensure smooth operation.
- Secure the stepper motor and linear carriage properly to minimize mechanical vibrations.
- Avoid to let go the stepper motor to the end of the axis because it can heat the instrument and can damage it.
- Temperature variations affecting sensor calibration.
- Electrical noise affecting sensor readings.
- Mechanical misalignment of the sensor during measurement.
- Effect of digitization of ADC on measured field properties.